TP3 Big Data – Ad Click-Through (CTR) Analysis

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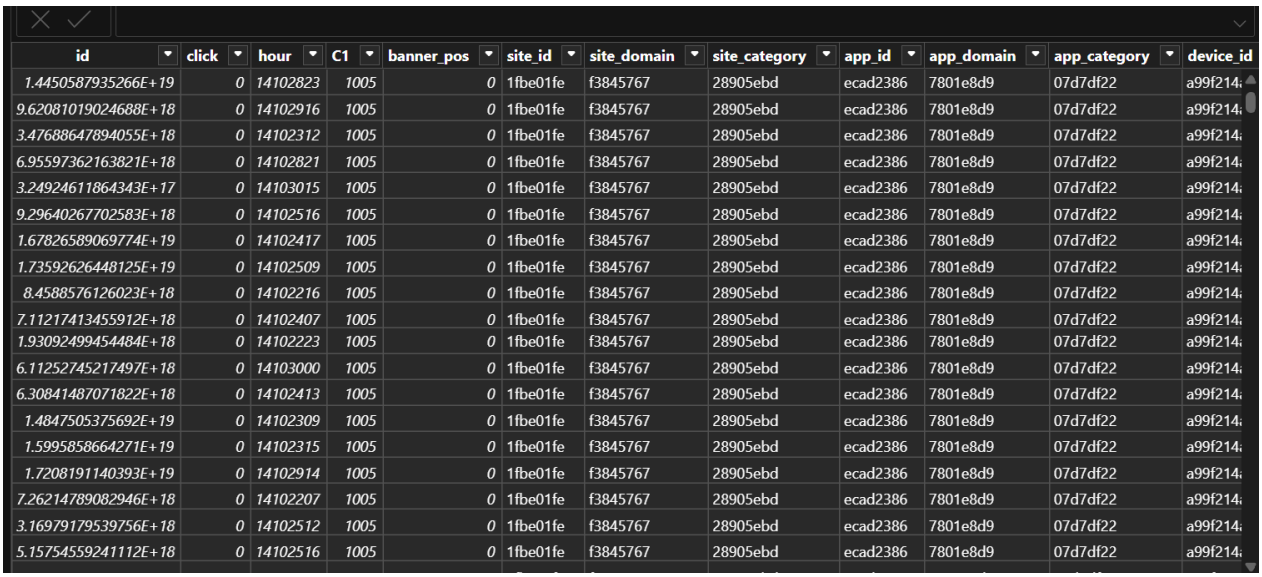
1 Dataset Description and Feature Selection

The CTR Prediction dataset, obtained from Kaggle, is used to predict whether a user will click on an online advertisement. It contains 24 columns describing advertising impressions, website and application information, device characteristics, and encoded advertising features. The target variable is `click`, which takes the value 1 when a user clicks on an advertisement and 0 otherwise.

For the prediction task, relevant features were selected, including `hour`, `banner_pos`, `site_id`, `site_category`, `app_id`, `app_category`, `device_model`, `device_type`, `device_conn_type`, and the numerical features `C14` to `C21`. These features provide information about the time of the impression, advertisement placement, website and application context, and user device characteristics. The `id` column was excluded because it primarily serves as a record identifier. Categorical features require appropriate encoding before being used by most machine learning algorithms. The selected features will be used to train and evaluate binary classification models for click-through prediction.

Data Type

Property	**Value**
Dataset	`ctrpredictiontrain` (ad impressions, CTR prediction)
Data type	Structured, tabular (one row = one ad impression)
Columns	24
Profiled rows	first 1000 (Power Query column profiling)
Target variable	`click` (binary: 1 = clicked, 0 = not clicked)
Tools	Power Query (cleaning), Power BI (visualization)



id	click	hour	C1	banner_pos	site_id	site_domain	site_category	app_id	app_domain	app_category	device_id
1.4450587935266E+19	0	14102823	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
9.62081019024688E+18	0	14102916	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
3.47688647894055E+18	0	14102312	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
6.95597362163821E+18	0	14102821	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
3.24924611864343E+17	0	14103015	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
9.29640267702583E+18	0	14102516	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.67826589069774E+19	0	14102417	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.73592626448125E+19	0	14102509	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
8.4588576126023E+18	0	14102216	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
7.11217413455912E+18	0	14102407	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.93092499454484E+18	0	14102223	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
6.11252745217497E+18	0	14103000	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
6.30841487071822E+18	0	14102413	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.4847505375692E+19	0	14102309	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.5995858664271E+19	0	14102315	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
1.7208191140393E+19	0	14102914	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
7.26214789082946E+18	0	14102207	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
3.16979179539756E+18	0	14102512	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214
5.15754559241112E+18	0	14102516	1005	0	1fbe01fe	f3845767	28905ebd	ecad2386	7801e8d9	07d7df22	a99f214

Figure 1: data

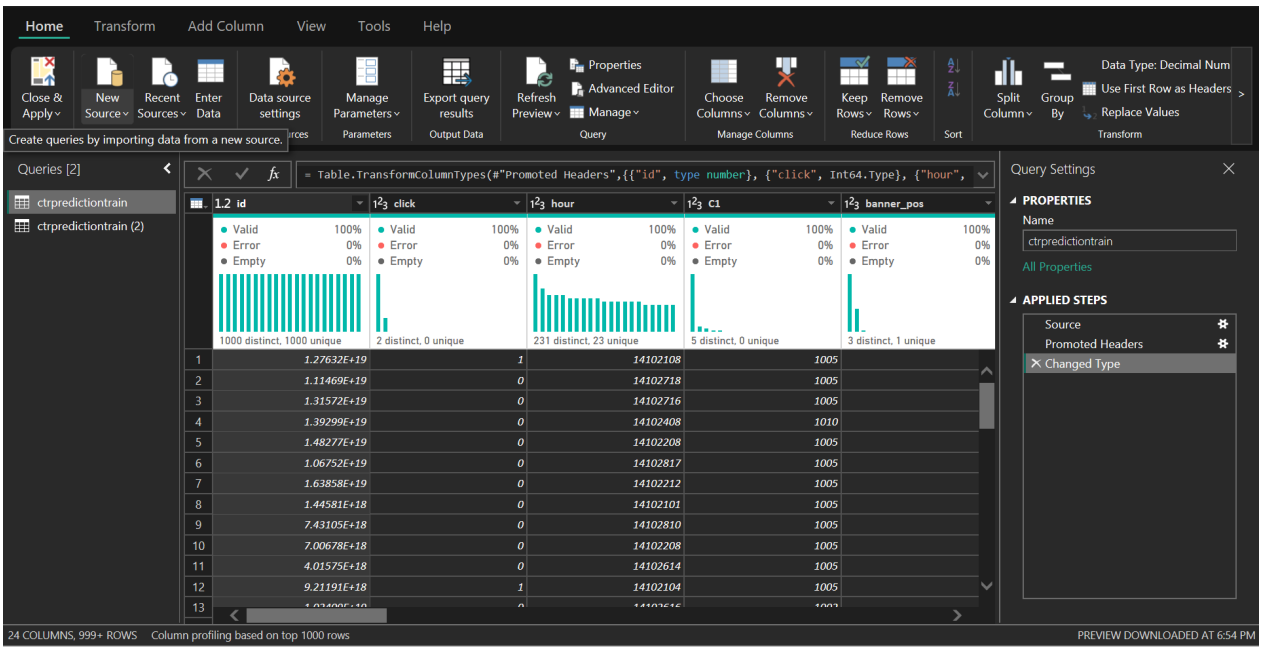


Figure 2: dataclean

2 Results

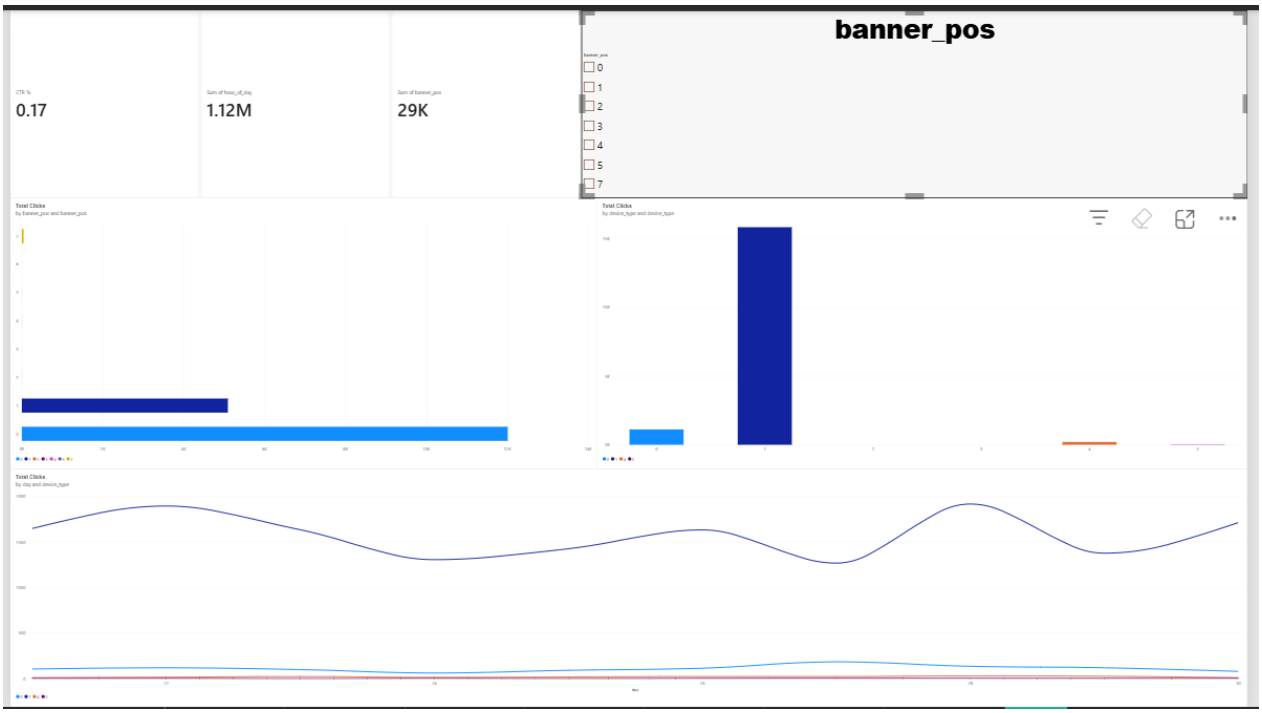


Figure 3: Overall Click Performance Across All Banner Positions

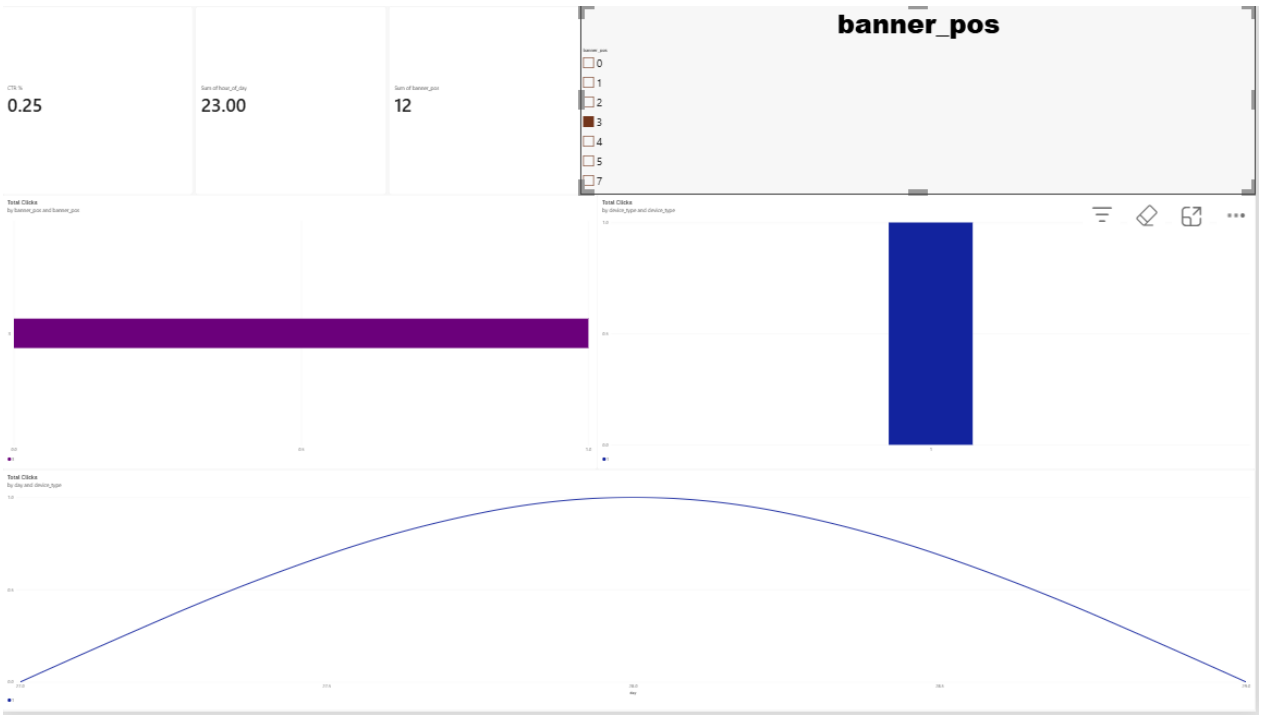


Figure 4: Banner Position 4: Click Performance (CTR 0.25%)

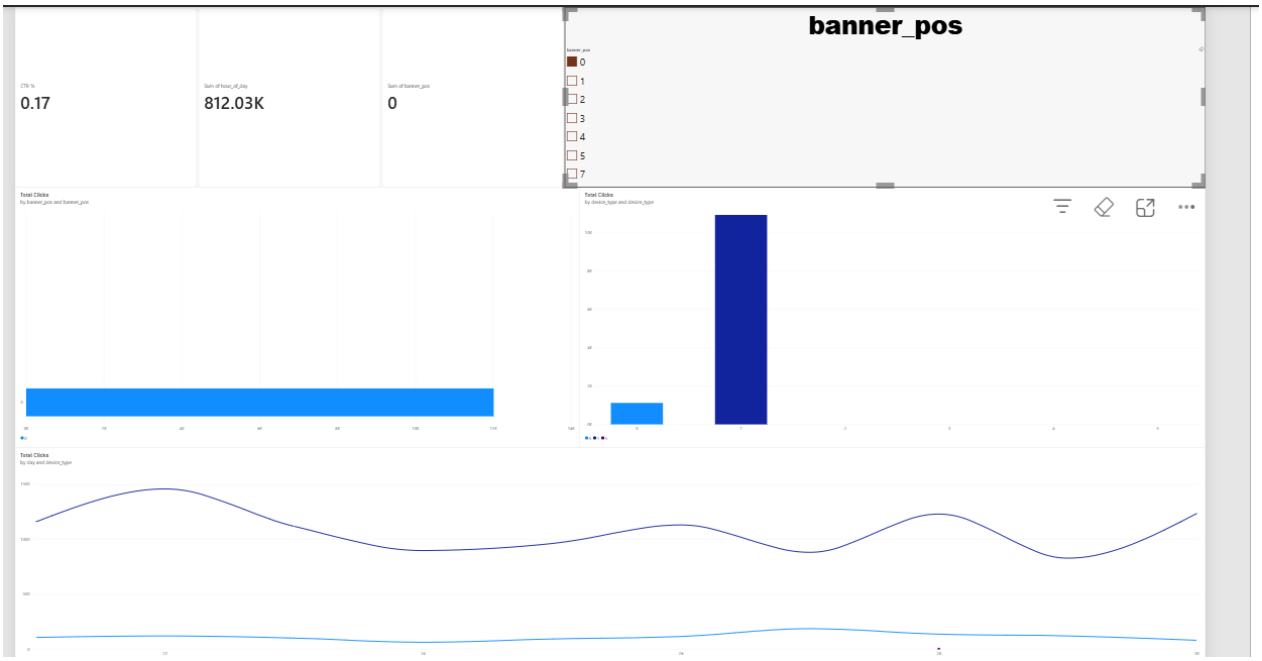


Figure 5: Banner Position 0: Click Performance (CTR 0.17%)